Attendance System with Facial Recognition

Abstract

Conventional attendance methods such as manual roll calls, RFID cards, and fingerprint-based systems are often time-consuming, inefficient, and vulnerable to proxy attendance. To overcome these limitations, this project presents a **Web-Based Attendance System with Facial Recognition** that automates the attendance process using computer vision and deep learning techniques.

The proposed system captures real-time facial images through a camera-enabled web interface and performs face detection and recognition using trained facial feature extraction models. Facial data is converted into numerical representations and compared with stored records to accurately identify individuals. Upon successful recognition, attendance is automatically recorded along with the corresponding date and time in a centralized database.

In addition to basic functionality, the system incorporates several practical improvements, including multi-angle face registration to enhance recognition accuracy, duplicate attendance prevention to ensure data integrity, and threshold-based matching to reduce false recognition. Basic liveness verification, such as face movement detection, is also included to minimize photo-based spoofing. Furthermore, a web-based administrative dashboard is provided for easy monitoring and report generation.

Overall, the proposed solution offers a reliable, contactless, and efficient attendance management system suitable for educational institutions and organizations.